

PALAKPREET KAUR

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OBJECTIVE

A Results-driven and Passionate B.Tech Computer Science (AI & ML) student with a strong foundation in Machine Learning, Data Analysis, and Python programming. Experienced in developing ML models, computer vision and Deep learning applications, and deploying solutions using Flask. Seeking internship opportunities to apply technical and analytical skills in real-world projects

WORK EXPERIENCE

Technical Trainee – Machine Learning

Jun 2025 – Aug 2025

Ziion Technology, Mohali, India

- Assisted in preprocessing and analyzing datasets of 10,000+ records for ML model training.
- Contributed to model development and evaluation improving performance metrics.
- Collaborated with a team of 4 members on real-world machine learning projects.

EDUCATION

B.Tech CSE (AI & ML)

2023 – 2027

DAV Institute of Engineering & Technology, Jalandhar

CGPA(till 5th sem):8.47

Senior Secondary (CBSE)

2022 - 2023

Jawahar Navodaya Vidyalaya

88%

Matriculation (CBSE)

2020 - 2021

Jawahar Navodaya Vidyalaya

92%

SKILLS

Programming: Python, C, C++, Java, MySQL, R

Libraries: Pandas, NumPy, Scikit-learn, OpenCV

Tools: Visual Studio Code, Jupyter, Flask, Streamlit, GitHub, TF-IDF, PyTesseract

AI & Machine Learning: Regression, Classification, Model Evaluation, LLMs, Neural Networks

Soft Skills: Communication, Teamwork, Leadership, Problem-Solving

PROJECTS

Brain Tumor Detection Using Deep Learning with Explainable AI(Research Project)(Ongoing):

- Developed an AI-based MRI analysis system for automated brain tumor classification using CNN (ResNet18).
- Built preprocessing pipelines for image resizing, normalization, and augmentation.
- Implemented Federated Learning using Flower and FedAvg for privacy-preserving model training.
- Integrated Explainable AI techniques (Grad-CAM & SHAP) for model interpretation.
- Classified MRI scans into Glioma, Meningioma, Pituitary Tumor, and No Tumor categories.

Self-Learning Chatbot:

- Developed an NLP-based chatbot using Python and Flask with TF-IDF for intelligent query matching.
- Implemented dynamic learning using SQLite database for storing and improving responses.
- Integrated Google Search API fallback mechanism for handling unknown queries.
- Applied Scikit-learn, TF-IDF Vectorizer, and Cosine Similarity for response generation.
- Designed an interactive chatbot interface using HTML, CSS, and Flask templates.

RESEARCH INTERESTS

- Deep Learning and Explainable AI for Medical Image Analysis
- Computer Vision (image classification, segmentation, and tumor detection)

- Natural Language Processing (NLP) and Conversational AI Systems
- Machine Learning (Regression, Classification, and Model Evaluation)
- Data Analysis, Visualization, and Performance Optimization
- Applied AI for Real-World Healthcare and Language Applications